MIL-P-0016208E(GL)

23 June 1975
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29 September 1969

MILITARY SPECIFICATION

PUNCHES AND DIES, GROMMET INSERTING

This limited coordination Military specification has been prepared by the U.S. Army Natick Development Center(GL), Natick, MA 01760, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-P-16208D. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers grommet inserting punches and dies.

* 1.2 Classification. The punches and dies shall be of the following types, classes and sizes, as specified (see 6.2).

Type I - Plain grommet inserting
Type II - Spur grommet inserting

Class - 1 with integral dies
Class - 2 with inserted dies

Sizes (see paragraph 3.3 and applicable figures)

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 5120

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part if this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

E 18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distrubuted among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Standard product. The punches and dies covered by this specification shall be the manufacturer's current commercial product except for any changes necessary to comply with this specification.

* 3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for punches and dies, provided the completed item complies with all provisions of this specification.

* 3.3 Design and construction. The type I, and type II, class 1 and 2 punches and dies and die holder (when applicable) shall be of same designated size, as specified (see 6.2). The finished surfaces of punches and dies shall be concentric within 0.008 inch. Punches, dies, and die holders shall conform to the design and construction specified herein, and as shown on the applicable figures as follows:

Type I punch and die - Figure 1 and figure 2

Type II, class 1, punch and die - Figure 3 and figure 4

Type II, class 2, punch, die and die holder - Figure 5 and figure 6

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* 3.3.1 Punches. Type I, and type II, class 1 and 2 punches shall be formed of steel. The handle portion of the punch may be made of malleable iron. The punch end shall have a hardness of Rockwell C 42-50 inclusive, or may be casehardened not less than 0.010 inch deep and have a superficial hardness of Rockwell 15N, 81.5, - 85.5, inclusive, when tested as specified in 4.3.1.

* 3.3.2 Die and die holders. Type I and type II, class 1 and 2 dies shall be made of steel. Die holders for type II, class 2 dies shall be made of cast iron with an inserted hardened die. Dies shall have a hardness of Rockwell C 42-50 inclusive, or type I, and type II, class 1 dies may be casehardened not less than 0.010 inch deep and have a superficial hardness of Rockwell 15N, 81.5, - 85.5 inclusive, when tested as specified in 4.3.1.

* 3.4 Finish. Working surfaces of punches and dies shall be polished to a surface roughness value of 63 microinches or better. Other surfaces shall be in accordance with the manufacturer's current commercial practice.

* 3.5 Marking. The punch and die shall be marked with "plain" or "spur" and the size designation, as applicable In addition, the manufacturer shall permanently and legibly indent his name, or trademark of such known character as to be easily identifiable to the manufacturer on the punch and die.

* 3.6 Workmanship. Punches and dies shall be free from defects such as cracks, pits, inclusion of foreign matter, burrs, rough or sharp edges. Working surfaces shall be free of tool marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification, or applicable purchase document.

* 4.2.2 End item inspection. A lot shall consist of all punches and dies of one type, class, and size offered for inspection at one time. The sample unit for this inspection shall be one complete punch and die.

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* 4.2.2.1 Visual examination. Examination shall be made of the punch and dies for defects set forth in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Visual examination of defects

Examine	Defect	Classification Major	Minor
Finish	Working surfaces not polished to 63 microinches or better	X	
Construction and workmanship	Crack, pit, burr, rough or sharp edge	X	
	Evidence of imbedded foreign matter		X
	Tool mark on working surface	X	
Marking	Missing, illegible, not as specified		X

* 4.2.2.2 Dimensional examination. Examination shall be made of the punch and die for defects in dimensions. Any dimension not within specification requirements shall be classified as a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units.

* 4.2.2.3 Testing of the end item. The sample unit shall be tested in accordance with 4.3.1. The inspection level shall be S-3 with an AQL of 2.5, expressed in terms of defects per hundred units.

* 4.2.3 Examination of preparation for delivery. Examination of preparation for delivery shall be in accordance with PPP-P-40.

* 4.3 Test.

* 4.3.1 Hardness test. The working surface of the punch and die shall be tested for hardness in accordance with ASTM E 18 to determine compliance with 3.3.1 and 3.3.2. Three readings shall be taken adjacent to the working surface of the punch and three readings shall be taken adjacent to the working surface of the die. The average of the readings shall be the hardness of the sample punch or die, providing each reading is within the specified range. Nonconformance to 3.3.1 or 3.3.2 constitutes failure of this test.

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5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, packing and marking. Preservation, packing, packaging and marking shall be in accordance with PPP-P-40. Levels of preservation, packaging and packing shall be as specified (see 6.2).

6. NOTES

* 6.1 Intended use. Punches and dies covered by this specification are intended for inserting grommets in duck, canvas, leather, rubber and similar materials. Type I punches and dies are intended for inserting plain grommets. Type II punches and dies are intended for inserting spur grommets.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Size of punch and die required (see 2.4).
(d) Selection of applicable levels of preservation, packaging and packing required (see 5.1).

6.3 The requirements for the type III punches and dies were deleted from this revision since the items are no longer catalogued.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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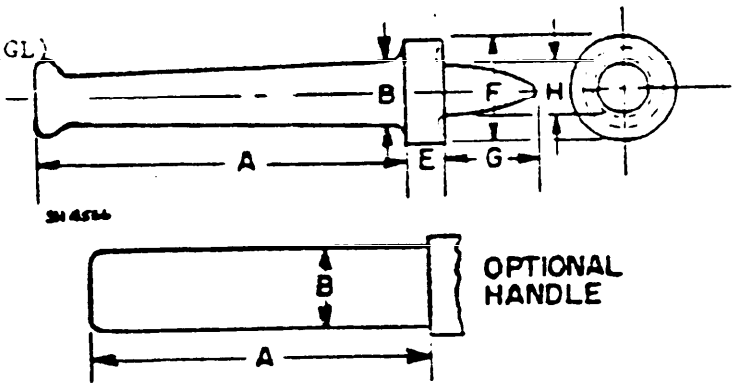
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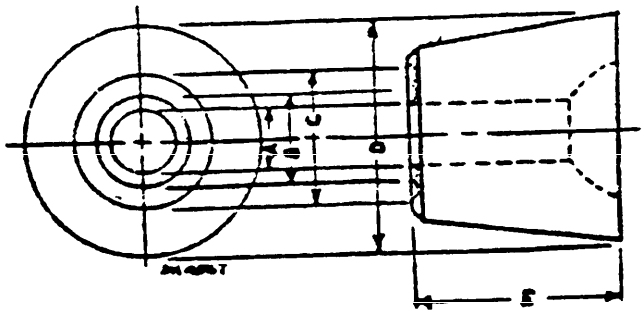
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Size Number	Handle		Punch end			
	Length	Diameter	Face		Fillet	
			Thickness	Diameter	Length	Major diameter
	A ± 1/4	B ± 1/16	E ± 1/8	F ± .025	G + 1/8 − 1/16	K − .006 + .010
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
00	3-1/8	7/16	3/8	.515	9/16	.182
0	3-1/8	7/16	3/8	41/64	3/4	.254
1	3-5/16	9/16	1/2	51/64	13/16	.296
2	3-11/16	9/16	17/32	29/32	1	.375
3	3-13/16	5/8	9/16	1	1-1/16	.438
4	3-7/8	5/8	9/16	1-1/8	1-1/8	.500
5	4-3/16	25/32	5/8	1-5/16	1-3/16	.625
6	4-1/8	1	3/4	1-47/64	1-3/4	.800
8	4-11/16	1-1/8	11/16	1-31/32	1-11/16	.915
10	4-5/8	1-5/32	13/16	2-31/64	2	1.253
11	4-1/2	1-15/64	27/32	3	2-1/16	1.555
15	5	1-25/64	7/8	3-35/64	1-15/16	1.980

Figure 1. Type I. Plain Grommet Inserting Punch

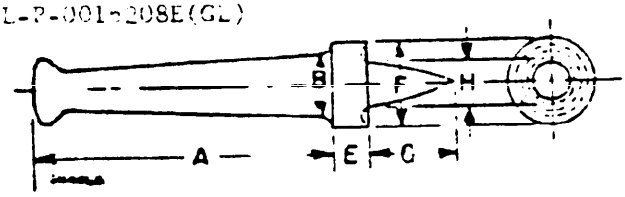
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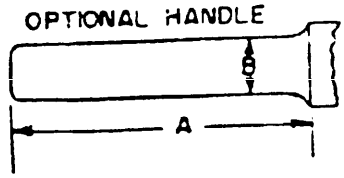


Size number	Die dimensions				Height of die
	Center hole	Inside bead	Outside bead	Overall	
	A +.010 −.005	B ± .020	C ± .005	D ± 1/4	E ± 1/4
	Inches	Inches	Inches	Inches	Inches
00	.157	———	.462	1-7/8	1-15/16
0	.254	.484	.553	1-7/8	1-15/16
1	.296	.634	.697	1-7/8	1-15/16
2	.390	.725	.822	1-7/8	1-15/16
3	.453	.874	.976	1-7/8	1-15/16
4	.500	.953	1.057	2-5/16	1-15/16
5	.625	1.125	1.223	2-5/16	1-15/16
6	.812	1.565	1.750	2-5/16	2-1/4
8	.937	1.406	1.937	2-15/16	2-1/4
10	1.281	1.750	2.375	3-5/16	3-3/8
11	1.578	2.187	2.812	3-1/2	3-1/4
15	2.046	2.562	3.312	4-3/16	3-9/16

Figure 2. Type I, Plain Grommet Inserting Die

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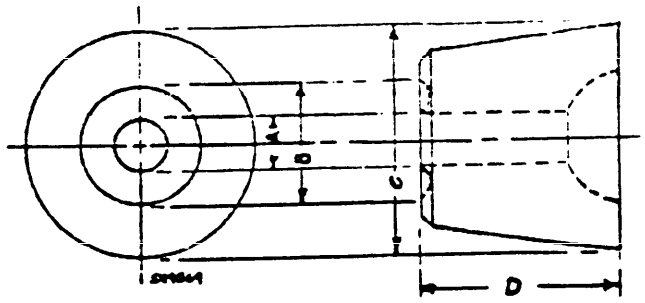




Size number	Handle		Punch end			
	Length	Diameter	Pass		Pilot	
			Thickness	Diameter	Length	Major diameter
	A ± 1/4	B ± 1/16	E ± 1/8	F ± .025	G +1/8 −1/16	H +.012 −.006
	Inches	Inches	Inches	Inches	Inches	Inches
0	3-1/4	5/8	7/16	23/32	3/4	.296
1	3-5/16	9/16	7/16	13/16	7/8	.375
2	3-11/16	9/16	1/2	15/16	1	.437
3	3-13/16	5/8	1/2	1-1/16	1-1/8	.468
4	3-7/8	5/8	1/2	1-1/4	1-1/8	.546
5	4-1/16	25/32	5/8	1-3/8	1-1/4	.625

Figure 3. Type II, Class 1, Intergral Die, Spur Grommet Inserting Punch

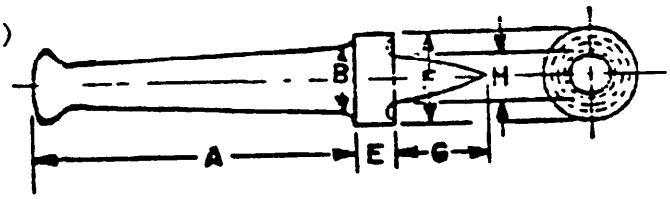
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Size number	Die diameters			Height of die
	Center hole	Band	Overall	
	A +.012 −.005	B ±.020	C ± 1/4	D ± 1/4
	Inches	Inches	Inches	Inches
0	.296	.750	1-7/8	1-15/16
1	.375	.875	1-7/8	1-15/16
2	.437	.937	1-7/8	1-15/16
3	.468	1.062	1-7/8	1-15/16
4	.546	1.218	2-5/16	1-15/16
5	.625	1.375	2-5/16	1-15/16

Figure 4. Type II, Class 1, Intergral Die, Spur Grommet Inserting Die

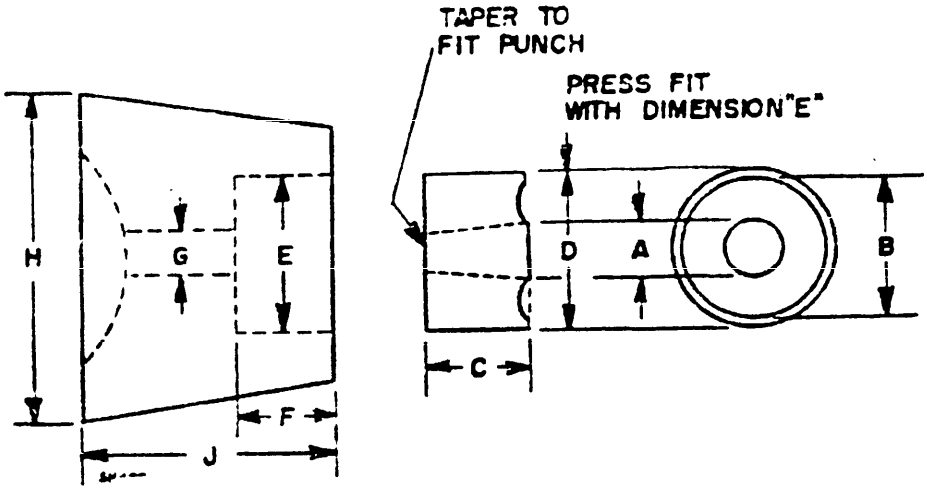
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Size number	Handle				Punch end			
	Length	Diameter			Face		Pilot	
	A nominal	B nominal	C	D	Thickness F (min.)	Diameter J ±0.005	Length G ±0.010	Major diameter M +0.000 −0.002
	Inches	Inch			Inch	Inches	Inches	Inch
0	3-5/8	3/4			1/2	3/4	13/16	0.296
1	4-1/8	3/4			5/8	15/16	1	.406
2	4-1/8	13/16			9/16	1	1-1/16	.437
3	4-1/4	7/8			9/16	1-1/8	1-1/16	.468
4	4-1/4	1			1/2	1-1/4	1-1/16	.578
5	4-3/8	1			7/16	1-7/16	———	.625

Figure 5. Type II, Class 2, Inserting Die Spur Grommet Inserting Punch

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Size number	Die insert dimensions				Block dimensions				
	Center hole	Bead	Length	O.D.	Insert cavity diameter	Insert cavity depth	Center hole	O.D.	Height
	A +0.005 −.000	B ±0.010	C ±1/32	D +0.001 −.000	E	F ± 1/32	G ± 1/32	H (min.)	J (min.)
	Inch	Inches	Inch	Inches	Inches	Inch	Inch	Inches	Inches
0	0.301	25/32	13/16	1	1	2/4	1/2	2	2
1	.411	25/32	13/16	1	1	3/4	1/2	2	2
2	.642	15/16	13/16	1	1	3/4	1/2	2	2
3	.473	1-1/16	13/16	1-1/4	1-1/4	3/4	1/2	2-3/8	2
4	.583	1-1/8	13/16	1-1/4	1-1/4	3/4	1/2	2-3/8	2
5	.630	1-3/8	13/16	1-3/8	1-3/8	3/4	1/2	2-3/8	2

Figure 6. Type II, Class 2, Inserted Die, Four Grommet Inserting Die

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